ClimateAdapt-Comfort

Decision-Support System for Passive Shelter Design

Across Indian Climate Zones

Research Question: Does a thermal-comfort model trained on global field data, coupled to building simulation, recommend shelter designs that reduce discomfort hours and energy demand, and does it beat PMV and adaptive-model baselines?

Python 3.13 | EnergyPlus v26.2 | XGBoost | NSGA-II | Streamlit

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1. Purpose & Research Question

The ClimateAdapt-Comfort project investigates whether empirical thermal comfort models trained on global field survey data, coupled to building thermal simulation and multi-objective optimization, can recommend passive shelter designs that minimize occupant discomfort hours and annual HVAC energy demand across distinct Indian climate zones.

The system processes data from the ASHRAE Global Thermal Comfort Database II (58,512 rows after complete-case filtering), simulates 10,000 building designs via EnergyPlus across 5 Indian cities, and uses NSGA-II multi-objective optimization to find Pareto-optimal shelter configurations.

2. Headline Results (Evidence-Gated)

All numbers are grounded strictly in literal command outputs recorded in docs/TRACKER.md and reports/evidence/.

S1: Comfort Model vs Adaptive Baseline - FAIL [E-012]

Target: Best ML model beats best baseline by >= +0.05 macro-F1 on identical complete-case test rows (58,512 rows) under Leave-One-Koppen-Group-Out CV.
Result: Physical Adaptive 90% achieved macro-F1 0.4939 (95% CI: [0.4907, 0.4971]); ML Champion (Ablation_A3_OutdoorTemp) achieved macro-F1 0.4785 (95% CI: [0.4754, 0.4816]).
Margin: -0.0154 (reported honestly as a negative result).

S2: Surrogate Model Accuracy - PARTIAL [E-019]

Target: XGBoost surrogate regressors reach R-squared >= 0.95 on held-out 20% test designs.
Result: 23 of 25 surrogate models achieved R-squared >= 0.95. Two failures: Mumbai O1_ml (R-squared = 0.9406) and Shimla warm_ml (R-squared = 0.9225). Mean R-squared on annual HVAC energy (O2) reached 0.987 across all 5 climate zones.

S3: Direct EnergyPlus Verification - PASS [E-020]

Target: At least one verified Pareto design per city beats BOTH R1 (Lightweight) and R2 (Conventional) reference shelters across all three objectives.
Result: Passed in all 5 climate zones (36 of 50 verified designs beat both reference shelters on all 3 objectives simultaneously).

City	Climate Zone	Designs Beat Both	Discomfort Reduction	Energy Reduction
Jodhpur	Hot-Dry	7 / 10	75.72% -> 36.64%	281.88 -> 41.40 kWh/m2/yr
Mumbai	Warm-Humid	7 / 10	65.95% -> 41.97%	199.99 -> 102.94 kWh/m2/yr
New Delhi	Composite	6 / 10	70.40% -> 39.35%	279.40 -> 50.40 kWh/m2/yr
Bengaluru	Temperate	8 / 10	59.93% -> 45.74%	66.94 -> 36.61 kWh/m2/yr
Shimla	Cold	8 / 10	77.36% -> 67.95%	279.54 -> 74.08 kWh/m2/yr

S4: Decision-Support Dashboard - PASS [E-021]

Streamlit dashboard runs locally via 'streamlit run src/app/app.py'. Automated pytest test suite (tests/test_app.py) passed all 6 UI tests in 5.38 seconds with 0 errors. Zero external network or paid API dependencies.

3. Applicability Envelope & Confidence Tiers

Across all five simulated Indian climate zones, the ML Champion comfort model operates below the 80% training envelope threshold (27.91% to 77.26% of hours inside the DB II 1st-99th percentile feature envelope), driven by severe monsoon relative humidity and dry desert conditions.

City	Climate	All Features In-Envelo	Adaptive Valid Hour	Dual-Judge Spearma	Confidence Tier
Mumbai	Warm-Humid	27.91%	100.00%	0.945	Medium
Bengaluru	Temperate	43.39%	100.00%	0.943	Medium
Shimla	Cold	77.26%	73.97%	0.868	Low
New Delhi	Composite	53.12%	89.86%	0.195	Low
Jodhpur	Hot-Dry	53.91%	86.85%	0.109	Low

4. Technology Stack

Category	Libraries / Tools
Core Scientific & Data	NumPy 2.2.6, Pandas 2.3.2, SciPy 1.15.3, PyArrow 21.0.0
ML & Optimization	Scikit-learn 1.7.1, XGBoost 3.4.1, Optuna 5.0.0, PyMOO 0.6.2, SHAP 0.53.0
Thermal Comfort	pythermalcomfort 4.6.1, EnergyPlus v26.2.0
Visualization & Dashboard	Matplotlib 3.10.6, Seaborn 0.13.2, Streamlit 1.36.0
Dev & Testing	Pytest 9.1.1

5. Pipeline Phases & Measured Runtimes

The project is organized into 7 sequential phases, each with defined acceptance gates. Total measured pipeline runtime is approximately 85 minutes, with EnergyPlus simulation being the primary bottleneck at ~67 minutes.

Stage	Script / Module	Description	Runtime
Clean & Splits	src/data/build_building_group_map.py src/data/make_splits.py	Complete-case filtering (58,512 rows) 5-fold GroupKFold + 4-fold LOGO	14.2 s
Physical Baselines	src/models/baselines.py	PMV, Adaptive 80%/90%, IMAC NV/MM 1,000 bootstrap CIs	82.5 s
ML Comfort Models	src/models/train.py	LogReg, RF, MLP, XGBoost Ablations A1-A3; Champion calibration	48.1 s
Simulation Batch	src/sim/batch.py	10,000 EnergyPlus runs 1,000 LHS designs x 5 cities x 2 modes	66.7 min

Stage	Script / Module	Description	Runtime
Coupling & Series	src/optimize/run_phase4a.py	Hourly extraction (43.69M records) Running mean, dual judge scoring	162.3 s
Surrogates Tuning	src/models/train_surrogates.py	Optuna 5-fold CV (40 trials x 25 models) 7,483 valid designs	10.7 min
Optimization & Verify	src/optimize/nsga2.py	NSGA-II (Pop=100, Gen=200, 5 cities) 50 direct EnergyPlus verifications	2.1 min
SHAP Explainability	src/models/explain.py	TreeExplainer global and local attributions across 25 surrogates	38.4 s

6. Repository Architecture

File / Directory	Purpose
AGENTS.md	Universal project rules and evidence gating contract
requirements.txt	Pinned Python package dependencies (Python 3.13)
run_all.ps1	Windows PowerShell orchestrator script
docs/	PROJECT_SPEC.md (single source of truth), PHASES.md (phase gates), TRACKER.md (evidence ledger)
data/raw/	Immutable raw source datasets (gitignored)
data/processed/	Derived parquets, splits, surrogates, metrics, and Pareto sets
models/	Serialized champion comfort model + 25 XGBoost surrogates (5 cities x 5 objectives)
reports/figures/	Generated Pareto scatter plots and SHAP importance figures
reports/evidence/	Literal command execution logs backing every claim
src/app/app.py	5-page interactive decision-support Streamlit dashboard
src/data/	Data cleaning, group mapping, and split generation
src/models/	Model training, baseline evaluation, surrogates, and SHAP
src/optimize/	Coupling engine, reference shelters, and NSGA-II optimizer
src/sim/	IDF rendering, LHS sampling, EnergyPlus runner, and output parser
src/run_all.py	Master end-to-end Python pipeline orchestrator
tests/	Automated test suites: app UI, coupling logic, data split integrity
tools/	Utility scripts: claims audit, batch runners, station pipeline

7. Raw Data Sources

All derived files under data/processed/ are generated deterministically. The raw source datasets must be placed in data/raw/:

ASHRAE Global Thermal Comfort Database II (v2.1.0)

Source: ASHRAE Database II GitHub Repository (CenterForTheBuiltEnvironment).
109,033 rows x 53 columns. Licensed under Open Database License (ODbL).

Indian Climate EPW Weather Files (TMYx / ISHRAE)

Source: Climate.OneBuilding.org
EPW files for Jodhpur, Mumbai, New Delhi, Bengaluru, and Shimla.

ICE Database Educational Version 5.0 (Embodied Carbon)

Source: Circular Ecology Inventory of Carbon and Energy.

Educational Use Only. Not used in runtime pipeline or simulation models.

8. Installation & Usage

Prerequisites

- Operating System: Windows 10/11 x64 (PowerShell 5.1 / 7.x)
- Python Version: Python 3.13.7 (or >= 3.11)
- EnergyPlus Engine: EnergyPlus Version 26.2.0

Installation Steps

```
1.  Clone the repository and navigate to the root directory
2.  pip install -r requirements.txt
3.  python -m pytest tests/ -v        (verify tests pass)
4.  streamlit run src/app/app.py       (launch dashboard)
```

Run Full Pipeline

```
python src/run_all.py           (full pipeline, ~85 min)
python src/run_all.py --smoke    (micro-sample smoke test, ~2.3 min)
```

9. Project Governance & Hard Rules

- Evidence Gating: Never state a number or result without literal command output behind it. Save outputs under reports/evidence/.
- Raw Data Immutability: Never edit, move, rename, or delete anything under data/raw/. All derived files go to data/processed/.
- Phase Discipline: Work only on the phase marked IN PROGRESS in docs/TRACKER.md.
- Locked Decisions: If a decision looks wrong, add a Proposal in TRACKER.md with evidence, then stop and ask the human.
- No Random Splits: Use only grouped splits defined in Phase 2 (data/processed/splits.json). Assert no building_id leakage.
- Free and Local Only: No paid APIs, no cloud credits, no accounts requiring payment.
- Reproducible: Fixed seed 42. All code in src/, run via documented commands.
- No Fabrication: Missing data stays missing unless the spec says how to handle it.
- Failures Reported Honestly: Failed runs, bad metrics, and negative results are reported as-is.

10. Licensing & Terms

- ASHRAE DB II: Open Database License (ODbL) - commercial and academic use permitted with attribution.
- EnergyPlus: US DOE / NREL modified BSD licence - open source.
- ICE Database v5.0: Educational Use Only - not used in runtime pipeline.